REVISION WORKSHEET

UNITED ARAB EMIRATES
MINISTRY OF EDUCATION

الإمارات العربية المتحدة
وزارة التربية والتعليم

AL KHALIDIYA BOYS SCHOOL

Cycle 3 (10048) | Dibba Al Hisn | Sharjah Eastern

Measurement Practice Questions

Grade 9 Advanced | Module 1 Lesson 3 | Measurement

Name: _____ Class: _____ Date: _____

Answer all 30 questions. Select one option for each MCQ. Show equations, working, units and explanations for written questions.

Useful equations and guidance: Percentage uncertainty = (absolute uncertainty/measured value) × 100; mean = sum of readings/number of readings.

Term 1 2026–2027 | Textbook alignment: uploaded Grade 9 book, printed pages 13–16.

P01. Accuracy describes:

A. Closeness of repeated readings only B. The number of trials
C. The size of the object D. Closeness to an accepted reference value

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P02. Precision describes:

A. Closeness of repeated readings to each other B. Closeness to the true value only
C. The absence of units D. The largest possible error

P03. Which change mainly reduces random uncertainty?

A. Ignoring unusual results without investigation B. Repeating readings and averaging
C. Changing units only D. Using an uncalibrated instrument

P04. A balance reads 0.20 g with nothing on it. This suggests:

A. Perfect accuracy B. An error in SI definitions
C. A zero error D. An independent variable

P05. Which action helps avoid parallax error?

A. Looking from the side B. Standing farther away at an angle
C. Rounding every reading to one digit D. Reading the scale at eye level perpendicular to it

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P06. A length is reported as (40 ± 1) cm. Calculate its percentage uncertainty.

A. 5 % B. 1.25 %
C. 2.5 % D. -2.5 %

P07. A scale has smallest divisions of 0.4 cm. Estimate the reading uncertainty as half a division.

A. 0.4 cm B. 0.1 cm
C. -0.2 cm D. 0.2 cm

P08. A standard length is 20 cm but an instrument reads 20.4 cm. Calculate the absolute measurement error.

A. 0.4 cm B. 0.8 cm
C. 0.2 cm D. -0.4 cm

P09. Three repeated measurements are 12, 14 and 16 s. Calculate their mean.

A. 28 s B. 14 s
C. 7 s D. -14 s

P10. Repeated readings are 22, 24 and 26 cm. Calculate the range of the readings.

A. 8 cm B. 2 cm
C. 4 cm D. -4 cm

P11. Define a measurement.

P12. Why should uncertainty be reported with a measurement?

P13. Distinguish random error from systematic error.

P14. Explain why a precise instrument can still give inaccurate results.

P15. What is instrument resolution?

P16. A length is reported as (60 ± 1.5) cm. Calculate its percentage uncertainty.

P17. A length is reported as (80 ± 2) cm. Calculate its percentage uncertainty.

P18. A scale has smallest divisions of 0.6 cm. Estimate the reading uncertainty as half a division.

P19. A scale has smallest divisions of 0.8 cm. Estimate the reading uncertainty as half a division.

P20. A standard length is 30 cm but an instrument reads 30.6 cm. Calculate the absolute measurement error.

P21. A standard length is 40 cm but an instrument reads 40.8 cm. Calculate the absolute measurement error.

P22. Three repeated measurements are 13, 15 and 17 s. Calculate their mean.

P23. Three repeated measurements are 14, 16 and 18 s. Calculate their mean.

P24. Repeated readings are 23, 25 and 27 cm. Calculate the range of the readings.

P25. Repeated readings are 24, 26 and 28 cm. Calculate the range of the readings.

P26. A length is (10.0 ± 0.2) cm. Does a second value of 10.1 cm agree within this uncertainty? Explain.

P27. A balance consistently reads 0.50 g too high. How should you improve the measurements?

P28. Describe how to measure the thickness of one sheet of paper more reliably with a ruler.

P29. Readings are 5.1, 5.1 and 5.2 cm when the accepted value is 6.0 cm. Discuss accuracy and precision.

P30. A student removes a reading only because it disagrees with the hypothesis. Evaluate the decision.